B.Sc. (Entire Computer Science)-I, Level - 4.5 UG Certificate Level							
Sem:	I						
Paper Category:	DSC1-1 **(Major)**						
Paper Name:	OOPs with C++-I	**Paper Code :** G06-0101					
Credit:	02	**Theory:**	2 Hrs./Week				
Marks:	**UA:**	30	**CA:**	20	**Total:**	50	

Course Objective:

Students will learn;

1. The basic programming and OOPs concepts
2. Creating C++ programs
3. Tokens, expressions and control structures in C++
4. Arranging same data systematically with arrays
5. Classes and objects in C++

Course Outcomes:

Upon successful completion of this course, students will be able to

1. Describe OOPs concepts
2. Use functions and pointers in your C++ program
3. Understand tokens, expressions, and control structures
4. Explain arrays and strings and create programs using them

Unit-I: Introduction to (Object Oriented Programming) OOP:	**No. of Lectures:15**	**Weightage: 8-12 Marks**

Introduction: Introduction to algorithm and flowchart, Introduction to OOP, Featuresof OOP's- Class, Object, Data Abstraction and encapsulation, Data hiding, Message passing, polymorphism, inheritance, delegation, Comparison between POP(Procedural Oriented Programming) and OOP, Advantages of OOP's, Application of OOP

Introduction to C++: History of C++, C++ basics(C++ tokens)- Keywords, identifiers, data types, constants,operators, special symbols, control flow statements- Decision and iterative statements, Types of Variables- Value, pointer and reference, Structure of C++ program, Basics Input and Output- cin, cout objects, Introduction to array, pointer and template, Function and its types, Default argument, Parameter passing methods, inline function

Unit-II: Classes and Objects:	No. of Lectures:15	Weightage: 18-22 Marks

Introduction to class and object, Defining class (class specification), Creating object, Access specifier (Visibility modes)-public, protected, private, Class members- data members, member function, Defining member function inside and outside the class, Static data members, static member functions, Pointer to object, Array of an object, Returning objects from functions, Passing object as a parameter by value, by pointer, by reference, Dynamic memory allocation (new, delete), Friend function and friend class, nesting of classes, Constructors, characteristics of a constructor, Types of constructor- default, parameterized and copy Constructor overloading, Constructor with default argument, Destructor, characteristics of destructor, Static polymorphism (Function and Operator overloading) , rules to overload operator, unary and binary operator overloading, overloading operator using member function and friend function, Type conversion (type casting)- implicit and explicit.

Reference books:

1. OOP in C++ – E-balagurusamy
2. Mastering C++-K. R. Venugopal
3. The Complete Reference C++-Herbert Schildt